Weekly Research Journal

Cartographer-Based Real-Time Vehicle Motion Estimation

Yiming Lu

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Abstract

This week implemented and validated a Cartographer-based ROS 2 estimator for whole-vehicle planar motion. The node reads the map-to-base_link transform and publishes speed, map-frame vx and vy, movement direction, and vehicle heading. Sensor rates were approximately 10 Hz for LiDAR and 50 Hz for IMU. Filter tuning moved from a 0.025 m/s deadband, which suppressed a genuine 0.0184 m/s motion, to a one-second window with a 0.005 m/s deadband. Manual motion tests produced speed responses up to approximately 0.1859 m/s. VESC comparison, gain calibration, synchronized logging, signed forward speed, powered remote-control testing, and sensor fusion were not completed in this reporting week.

1 Weekly contribution

The work advanced from sensor and transform verification to a reusable real-time motion interface. Table 1 distinguishes the implemented estimator from later comparison and fusion tasks.

Table 1: Evidence and scope of the 15 October results.

Contribution	Evidence	Scope
Cartographer pose	Continuously updated map-to-base_link transform	Completed
Motion-state node	Five ROS 2 topics for velocity and orientation	Completed
Manual validation	Slow and faster translations up to about 0.1859 m/s	Functional test
VESC comparison/fusion	Proposed as the next experiment	Not completed

2 System configuration and estimator interface

After the vehicle was powered on, USB identity was verified before launching the sensor stack. Hokuyo LiDAR data were available at approximately 10 Hz and the VESC IMU stream at approximately 50 Hz. Static base_link-to-laser and base_link-to-imu transforms were present. Cartographer then provided the map-to-base_link pose used by the estimator.

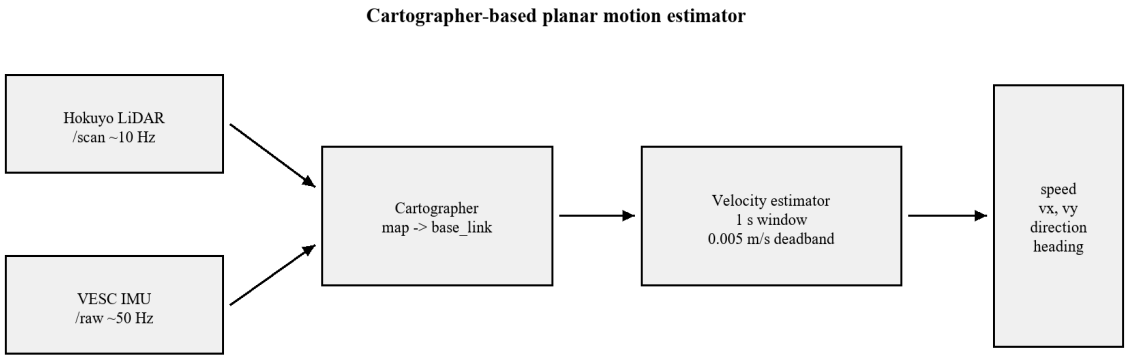


Figure 1: Cartographer-based motion-estimation architecture completed during the week. All outputs shown at right were verified in ROS 2.

Table 2: Real-time Cartographer motion interface on 15 October.

Topic	Physical quantity	Status
/cartographer_velocity/speed	Speed magnitude	Working
/cartographer_velocity/vx	Map-frame x velocity	Working
/cartographer_velocity/vy	Map-frame y velocity	Working
/cartographer_velocity/movement_direction	Direction of velocity vector	Working
/cartographer_velocity/heading	Vehicle yaw from Cartographer	Working

The later /cartographer_velocity/forward_speed topic is intentionally absent from Table 2: the signed body-axis projection was added after this reporting period and is not claimed as a 15 October result.

3 Estimation method and filter tuning

3.1 Velocity and direction from pose differences

For two Cartographer samples (x_1, y_1, t_1) and (x_2, y_2, t_2), map-frame velocity is estimated by finite differences:

$$v_x = (x_2 - x_1)/(t_2 - t_1), \quad v_y = (y_2 - y_1)/(t_2 - t_1). \quad (1)$$

$$v = \mathrm{sqrt}(v_x^2 + v_y^2), \quad \mathrm{phi} = \mathrm{atan2}(v_y, v_x). \quad (2)$$

The Cartographer yaw provides heading theta. Movement direction phi describes the velocity vector in the map frame, whereas heading theta describes body orientation. The quantities are related during ordinary rolling but are not definitionally identical.

3.2 Noise and deadband selection

Stationary localization jitter initially appeared as speeds of roughly 0.010 to 0.020 m/s. A temporary 0.025 m/s deadband removed these spikes, but a direct transform check showed a real displacement of about 0.459 m over 25 s, or 0.0184 m/s. Because 0.0184 is below 0.025, that threshold also erased genuine slow motion. The retained configuration used a one-second window and a 0.005 m/s deadband.

$$v = 0.459 \text{ m} / 25 \text{ s} = 0.0184 \text{ m/s}. \quad (3)$$

Table 3: Filter evolution during the week.

Configuration	Observed effect	Decision
0.5 s; 0.010 m/s	Stationary spikes remained	Reject
0.5 s; 0.025 m/s	Stationary zero, but 0.0184 m/s suppressed	Reject
1.0 s; 0.005 m/s	Retained slow motion with additional averaging	Adopt

4 Manual experimental validation

Manual translation was used to test the estimator before powered operation. Speed increased during motion and returned to zero after stopping. Reported values included 0.0449, 0.0734, 0.1264, and 0.1859 m/s. Predominantly straight motion produced a much larger vx component than vy. The plotted sample order is representative rather than a synchronized time axis.

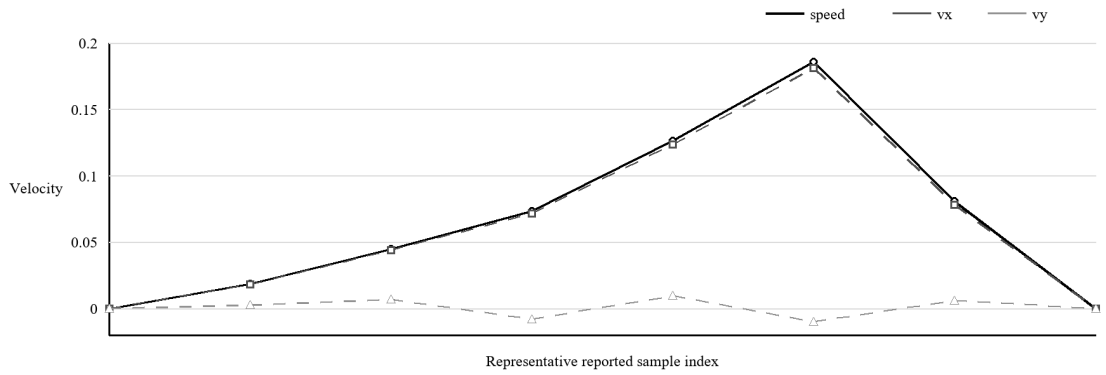


Figure 2: Representative reported manual-motion values. The plot demonstrates component response and return to rest; it is not a ground-truth speed trace.

Table 4: Interpretation of the manual validation evidence.

Observation	Supported conclusion	Not supported
Speed rose and returned to zero	Node responds to translation and stopping	Absolute accuracy
vx dominated during straight motion	Map-frame decomposition is physically plausible	Wheel-slip estimate
Heading remained available	Orientation output is integrated with velocity topics	Calibrated yaw error
Topics updated in real time	Interface is operational	Synchronized VESC residuals

5 Limitations

The manual tests do not provide independent distance-time ground truth. The velocity samples therefore demonstrate functionality and qualitative consistency only. The unusual roll values seen in one transform sequence also motivate a later review of frame orientation, although yaw remained sufficiently stable for the planar test.

6 Completed work and next steps

Table 5: Project status at the end of the reporting week.

Status	Item
Completed	Verified LiDAR, IMU, static transforms, and Cartographer pose.
Completed	Published speed, vx, vy, movement direction, and heading.
Completed	Adopted a one-second window and 0.005 m/s deadband.
Completed	Validated real-time response by manual translation and stopping.
Pending	Implement signed body-axis forward speed.
Pending	Conduct powered remote-controlled field testing.
Pending	Record timestamp-aligned Cartographer and VESC streams.
Pending	Calibrate VESC gain and evaluate fusion only after residual analysis.

7 Discussion and next experiment

The main result is an operational whole-vehicle motion estimator derived from Cartographer pose rather than motor rotation. The sequence of filter tests also shows why stationary-noise suppression must be balanced against slow-motion sensitivity. The final one-second window and 0.005 m/s deadband represent an experimentally motivated compromise for this platform.

The next experiment should add signed forward-speed projection, drive the vehicle under remote control, and log Cartographer and VESC outputs with shared ROS timestamps. VESC sign and scale should then be calibrated using measured distance and time. Sensor fusion remains a future design decision, not a completed result of this week.

Planned validation sequence

Step	Measurement	Decision enabled
1	Signed forward-speed projection	Separate forward and reverse motion
2	Powered field run	Test estimator under actual driving
3	Synchronized VESC and Cartographer log	Estimate sign, delay, and residuals
4	Measured-distance ground truth	Calibrate scale before fusion